Maynard 2020 LUAD — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/38 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands: an integration sweep (θ 5→40) read on the embedding gives one connected component at θ=20 (iLISI median 1.00→2.00 of 7); θ=40 over-integrates into detached islands. Both poles (Body / Terminus) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

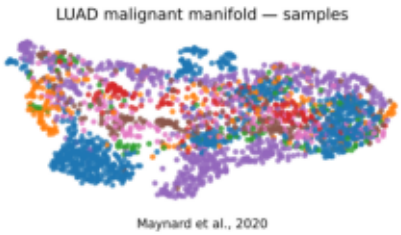
Gast and pSom co-vary (r=+0.43) at one origin pole; both oppose Crest (r=−0.74 / −0.64). Along pseudotime Gast/pSom decline (ρ=−0.64 / −0.63) and Crest rises to a terminal readout (ρ=+0.79; Crest-high decile mean pt=0.83). Joint pSom&Crest-high depleted (enrichment 0.29) — an inverse gradient, not a co-expression hub; Crest enriches the pSom/Gast-low pole, so no inversion flag.

Gate 3 — cross-repression → NA

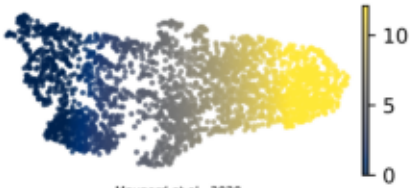
Single monotone axis — 0 branch nodes (curve & PPT tree), fold-free (self-approach 2.96), and no Gast/pSom-overlap hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

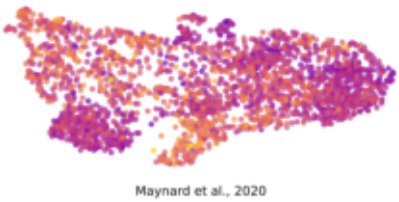
Manifold — samples



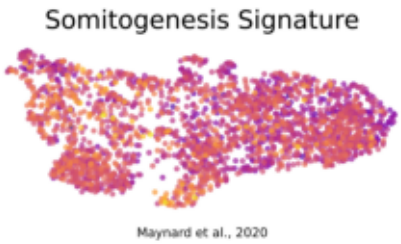
Pseudotime



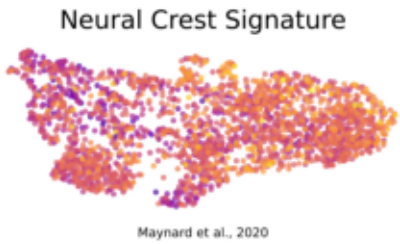
Gastrulation



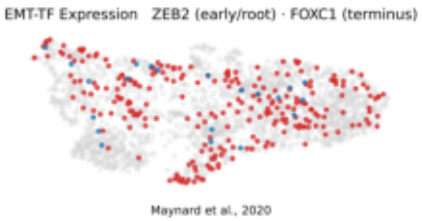
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
LUAD (Maynard et al., 2020), SmartSeq2 — 50 samples / 19,777 cells (master scope); malignant only: 3,184 cells; after ≥100-cell thresholding 2,486 cells across 7 tumour-bearing samples and island removal 2,396 cells enter the trajectory (whole manifold, no scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Maynard's metastatic-LUAD malignant manifold is a single monotone Gast/pSom → terminal-neural-crest gradient (somitogenesis-high origin → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The published cell-cycle annotation is NaN for malignant cells, so cycling was scored de novo; it is diffuse (22%), the modules are cell-cycle-free (Gast 0/81, Crest 0/104 cell-cycle genes), pSom↔Crest anti-correlation survives in non-cycling cells (r=−0.64→−0.67) and pseudotime↔cycling co-variation is negligible (ρ=−0.04) — the axis is developmental, not proliferative. FOXC1 (42% expressing, ρ=+0.15) is the robust neural-crest terminus factor; the root end is weakly marked — only ZEB2 (~3%, at the detection floor) surfaces as an early factor and its call is run-sensitive.

Parameter choices

Min cells/sample	≥100 floor (7/38; gap non-dominant)	**Integration**	Harmony θ=20; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Poles**	Body 1,198 / Terminus 1,198
Geometry	Linear (non-branching); 0 branch nodes	**EMT-TFs**	ZEB2 (early/root) · FOXC1 (terminus, 42%)